कोशी २०८१
प्रदेश लोक सेवा आयोग, कोशी प्रदेश, विराटनगर
प्रदेश निजामती तथा स्थानीय सरकारी सेवा अन्तर्गत प्राविधिक तर्फ इन्जिनियरिङ्ग सेवा, सिभिल समूह, जनरल उपसमूह, अधिकृतस्तर सातौं तह, इन्जिनियर वा सो सरह पदको खुला प्रतियोगितात्मक लिखित परीक्षा
विषय: सेवा सम्बन्धी | पत्र: द्वितीय | पूर्णाङ्क: १०० | समय: ३ घण्टा | मिति: २०८१।११।०७

Section-A (25 Marks)

1. **Explain the factors influencing the selection of bridge types in Nepal. [5]**

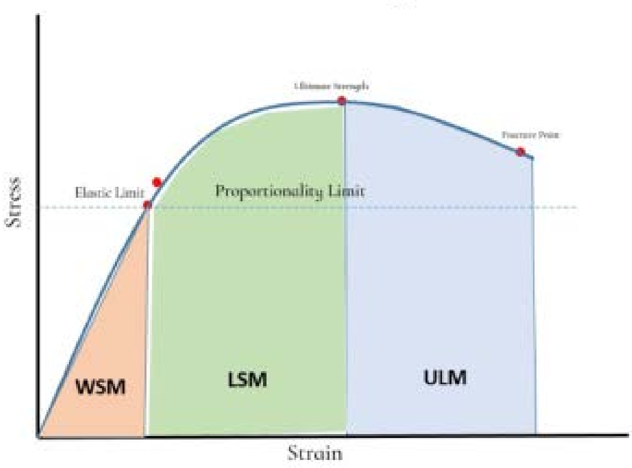
Answer:
The selection of an appropriate bridge type in Nepal is a complex decision influenced by the country's unique topography and socio-economic conditions. The key factors include:

i. **Topography and Geomorphology:** In the steep Himalayan and Hill regions, long-span suspension or arch bridges are preferred to avoid piers in deep gorges. In the Terai, multi-span RCC beam and slab or truss bridges are common due to wide, shallow riverbeds.
ii. **Hydrology and Scour Depth:** High-velocity Himalayan rivers with high sediment loads require bridge types that minimize obstruction to flow. The potential for Glacial Lake Outburst Floods (GLOF) necessitates higher freeboard and robust pier protection.
iii. **Geotechnical Conditions:** The availability of firm rock foundations in hills allows for arch or cantilever bridges, while deep alluvial deposits in Terai require deep foundations (wells or piles).
iv. **Seismicity:** Nepal lies in a high-intensity seismic zone. Bridge types with high ductility and lower self-weight (like steel trusses) are often prioritized for better seismic performance.
v. **Logistics and Accessibility:** In remote areas with limited road access, "Trail Bridges" or "Pre-fabricated Steel Trusses" are selected because components can be transported by porters or mules and assembled on-site.
vi. **Economic Viability:** The "Life Cycle Cost," including initial construction and long-term maintenance, is considered. RCC bridges are often favored for lower maintenance despite higher initial costs compared to timber.

2. **Discuss the key principles, advantages and limitations of working stress philosophy and the limit state philosophy in the design of reinforced concrete structures. Provide examples of their applications in the context of Nepal's construction industry. [4+4+2=10]**

Answer:

a. **Working Stress Method (WSM):**
- **Principles:** Assumes concrete and steel behave linearly elastic. Stresses under service loads are kept below "Permissible Stresses," which are a fraction of the ultimate strength ($Stress = Strength/FOS$). It uses the Modular Ratio (m) to relate steel and concrete.
- **Advantages:** Simple calculations; ensures high serviceability (low cracking/deflection) due to conservative stress levels.
- **Limitations:** Leads to uneconomical (over-designed) sections; does not account for the non-linear behavior of concrete at high loads or the actual factor of safety against collapse.



b. **Limit State Method (LSM):**

- **Principles:** A non-deterministic approach that considers "Limit States" beyond which the structure ceases to perform. It uses partial safety factors for both loads (γ_f) and materials (γ_m). Key states are Limit State of Collapse (Strength) and Limit State of Serviceability (Deflection/Cracking).
- **Advantages:** More economical and realistic design; accounts for uncertainties in loading and material strengths separately; ensures a uniform safety margin.
- **Limitations:** Complex mathematical modeling; requires stringent quality control on-site to ensure material strengths match design assumptions.

c. **Applications in Nepal:**

- **WSM Application:** Still commonly used in Nepal for the design of **Water Retaining Structures** (Water tanks, intake wells) to ensure they remain crack-free (water-tight). It is also the basis for many older bridges.
- **LSM Application:** The standard practice for modern **High-rise buildings** in Kathmandu and current **Bridge Designs** to optimize material use and enhance seismic ductility.

3. **What are the typical factors that affect the selection of foundation? Should foundation be differed as per the geographical conditions of Nepal? Suggest typical foundation that should be designed as a prototype in Terai and Himal with reasons? [5+2+3=10]**

Answer:

a. **Factors Affecting Foundation Selection:**

- **Load Magnitude:** Type and intensity of the superstructure load (Dead, Live, Wind, Seismic).
- **Soil Bearing Capacity:** The strength of the underlying strata.
- **Groundwater Table:** High GWT affects bearing capacity and requires dewatering or specialized foundations.
- **Settlement Sensitivity:** Permissible total and differential settlement of the structure.
- **Depth of Scour:** Critical for bridges and hydraulic structures in Nepal.
- **Adjoining Structures:** Proximity to existing buildings.

b. **Geographical Differentiation in Nepal:**

Yes, foundations must differ significantly because Nepal's geography ranges from deep alluvial plains (Terai) to unstable mountain slopes and rocky peaks (Himal). A "one-size-fits-all" approach leads to either structural failure or extreme uneconomic design.

c. **Prototype Suggestions:**

- **Terai Region (Prototype: Deep Foundations - Piles/Wells):**
  - *Reason:* Terai consists of deep layers of silt and sand with high GWT. Shallow footings often face settlement issues or liquefaction during earthquakes. Piles bypass weak surface layers to reach firmer strata.
- **Himal/Hill Region (Prototype: Shallow Foundations - Isolated/Raft on Rock):**
  - *Reason:* Firm bedrock is often available at shallow depths. Isolated footings or "Stepped Footings" are used to accommodate slope gradients. The primary challenge here is slope stability and preventing differential settlement on weathered rock.

Section-B (25 Marks)

4. **Briefly explain the water resource use priority as per Water Resource Regulation, 2050. [5]**

Answer:

According to the Water Resources Act, 2049 and Regulation, 2050 of Nepal, the priority for utilizing water resources is legally mandated as follows (in descending order):

i. **Drinking water and domestic users:** Highest priority for human survival.
ii. **Irrigation:** For food security and agricultural development.
iii. **Agricultural Uses:** Such as animal husbandry and fisheries.
iv. **Hydro-electricity:** For energy production.
v. **Cottage Industry, Industrial enterprises, and Mining:** For economic production.
vi. **Navigation:** For transport.
vii. **Recreational Uses:** Such as tourism and water sports.
viii. **Other uses:** Any miscellaneous uses not mentioned above.

Key Principle: No person is allowed to utilize water resources in a way that causes substantial adverse effects on the environment or downstream users, regardless of priority.

5. **Nepal is characterized by diverse topography and climatic conditions. Discuss the challenges and considerations associated with accurate rainfall measurements and related analysis in this context. Provide examples of how these challenges impact flood forecasting in Nepal. [7+3=10]**

Answer:

A. Challenges and Considerations:

- **Topographic Variation (Orographic Effect):** Rainfall in Nepal varies drastically over short distances due to altitude. A rain gauge in a valley may not represent the heavy precipitation on a nearby ridge.
- **Network Density:** The number of functional, automatic weather stations (AWS) is low, especially in high-altitude ungauged catchments.
- **Data Consistency:** Historical records often have "gaps" due to station maintenance issues or political instability. Consistency must be checked using **Double Mass Curve** analysis.
- **Inaccessible Terrain:** The rugged, high-altitude landscape makes installing and maintaining a dense network of automatic weather stations (AWS) logistically difficult and expensive.
- **Measurement Inaccuracies:** High wind speeds at high altitudes (25–30 m/s) and freezing temperatures can cause significant catch errors in standard rain gauges.
- **Extreme Events:** Standard gauges often fail or overflow during extreme cloudbursts (e.g., 2024 monsoon), leading to underestimated peak data.
- **Accessibility:** Manual gauges in remote areas lead to delays in data collection and human errors in reading.
- **Climatic Variability:** More than 80% of rainfall occurs during the monsoon (June–September), with annual amounts ranging drastically from 150mm to over 5000mm. Recent trends show more intense rainfall over shorter periods, complicating historical trend analysis.

B. Impact on Flood Forecasting in Nepal:

- **Underestimation of GLOF/Flash Floods:** Lack of real-time rainfall data in the high Himalayas makes it impossible to provide early warnings for flash floods in rivers like the Bhotekoshi or Melamchi. If localized extreme rainfall (cloudbursts) is missed by the gauge network, hydrological models will under-predict the peak runoff, leading to inadequate flood warnings.
- **Inaccurate Lead Time:** Inaccurate "Time of Concentration" analysis due to poor rainfall intensity data reduces the window for evacuation in the Terai plains.
- **Model Bias:** Flood forecasting models (like those used by DHM) rely on "Representative Rainfall." If the gauge network is sparse, the model may predict a minor rise while a devastating flood occurs (e.g., West Rapti floods).
- **Compounding Hazards:** Rainfall triggers landslides that can block rivers, leading to landslide-induced floods or Glacial Lake Outburst Floods (GLOFs). These secondary hazards are harder to forecast based on rainfall data alone.

6. **What are the major problems in operation and management of irrigation systems of Nepal? Describe the pros and cons of peaking ROR storage. [5+5=10]**

Answer:

a. **Major Problems in Irrigation O&M:**
   - **Siltation:** High sediment load in Nepalese rivers chokes canals and damages structures.
   - **Poor Infrastructure:** Many systems (FMIS) have temporary headworks that are washed away every monsoon.
   - **Resource Gap:** Insufficient government budget for periodic maintenance and lack of "Internal Resource Generation" by Water User Associations (WUA).
   - **Water Conflicts:** Inequality in water distribution between "Head-reach" and "Tail-end" farmers.
   - **Changing Climate:** Shift in rainfall patterns makes traditional "Duty and Delta" calculations obsolete.

b. **Peaking ROR (PROR) Storage:**

PROR plants use a small pond (Daily Pondage) to store water during off-peak hours to run at full capacity during peak demand (e.g., 4-6 hours in the evening).

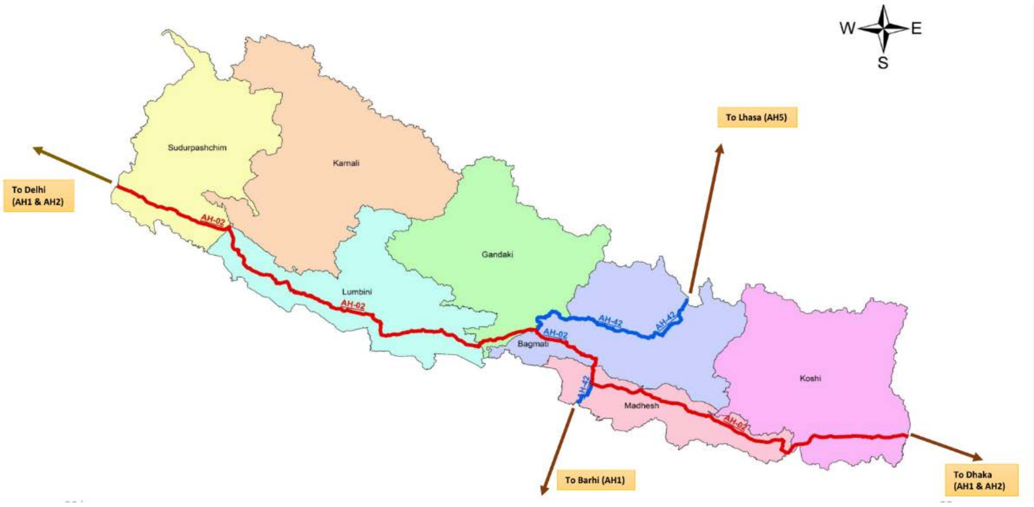
- **Pros:**
  - **Grid Stability:** Helps meet the high evening peak demand in Nepal.
  - **Revenue:** Energy generated during peak hours has a higher tariff (PPA).
  - **Flexibility:** Faster response compared to pure ROR or large Storage plants.
- **Cons:**
  - **Environmental Impact:** Diurnal fluctuation in downstream water levels disrupts aquatic life (Ecological flow issues).
  - **Cost:** Higher capital cost due to the requirement of a larger headrace/desanding basin and a pondage area.
  - **Sedimentation:** The pond acts as a silt trap, requiring frequent flushing.

Section-C (25 Marks)

7. **Highlight the significance of the Asian Highway in Nepal's transportation system. [5]**

Answer:

The Asian Highway (AH) network, specifically AH2 (East-West Highway) and AH42 (Lhasa-Kathmandu-Delhi), is vital for Nepal:



- **International Connectivity:** Connects landlocked Nepal to regional seaports (Haldia/Visakhapatnam) and neighboring markets (India and China).
- **Trade Facilitation:** Standardizes road geometry (width, gradients, pavement) to handle heavy international freight containers, reducing transit time and costs.
- **Economic Corridors:** Promotes the development of Special Economic Zones (SEZs) and industrial clusters along the highway (e.g., Bhairahawa, Birgunj).
- **Tourism:** Enhances "Cross-border Tourism" by providing high-quality road access to Buddhist circuits (Lumbini) and Himalayan gateways.
- **Standardization:** Forces Nepal to upgrade its national standards to meet international AH standards (like Kathmandu-Terai/Madhesh Fast Track Road is of AH standard), improving overall road safety and durability.

8. **Discuss the challenges and design considerations specific to hill roads in Nepal, covering alignment selection, gradient determination and the integration of retaining and slope protection structures. [10]**

Answer:

Constructing roads in Nepal’s hilly terrain requires a balance between engineering stability, economic feasibility, and environmental preservation.

Alignment Selection

- **Ridge vs. Valley Alignment:** Ridge alignments are often preferred because they require fewer cross-drainage structures and are less prone to flooding. However, valley alignments may be necessary to connect major settlements.
- **Geological Stability:** Alignment must avoid active landslide zones, weak rock formations, and adverse bedding planes. "Hairpin bends" are used to gain elevation but must be minimized to avoid excessive tire wear and traffic congestion.

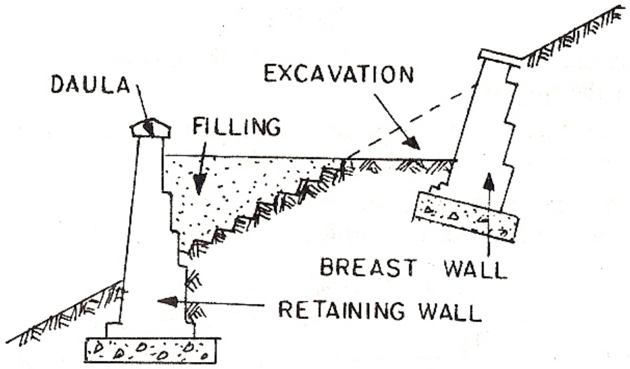
- **Environmental Sensitivity:** Alignments should minimize the "cutting and filling" of slopes to prevent destabilizing the natural hillside and causing erosion.

Gradient Determination

- **Ruling Gradient:** In hill roads, the ruling gradient is typically kept around **5% to 7%** to ensure heavy vehicles can climb safely without overheating.
- **Limiting and Exceptional Gradients:** Steep gradients (up to 10% or more) are used in short stretches for rapid elevation gain but require extra widening on curves and better surface friction.
- **Grade Compensation:** On sharp horizontal curves, the gradient must be reduced (compensated) to offset the additional tractive effort required by vehicles turning.

Retaining and Slope Protection Structures

- **Retaining Walls:** Used on the "valley side" to support the road embankment and prevent the road from sliding down the slope.



- **Breast Walls:** Constructed on the "hill side" to prevent soil and rock fragments from falling onto the roadway (slope stabilization).
- **Drainage Integration:** Proper catch drains and chutes are essential; water trapped behind a retaining wall increases pore water pressure, which is a primary cause of wall failure in Nepal.
- **Bio-Engineering:** Techniques such as planting deep-rooted grasses (e.g., Vetiver), brush layering, and palisades are integrated with civil structures to provide long-term surface protection against erosion and shallow landslides.

9. **Describes the types of road maintenance practiced in Nepal. Highlight its short coming and recommend suitable measures. [6+4=10]**

Answer:

a. Types of Maintenance (as per DoR):

- **Routine Maintenance:** Continuous activities like cleaning drains, cutting grass, and filling small potholes.
- **Periodic Maintenance:** Large-scale works performed every 4-6 years, such as "Resealing" or "Overlaying" to restore the riding surface.
- **Recurrent Maintenance:** Performed several times a year (e.g., patching, edge repair).
- **Emergency/Extraordinary Maintenance:** Immediate repair of landslide blocks or bridge washouts during monsoon.

b. Shortcomings:

- **Reactive approach:** Maintenance is often done only after the road is severely damaged (Failure-based).
- **Insufficient Funding:** Budget allocated to the Roads Board Nepal (RBN-a self-governing, self-sustaining entity established under the Roads Board Act 2058 (2002) with the primary goal of providing a sustainable fund for the planned maintenance of the country's roads) is often inadequate for the expanding network.
- **Poor Quality Control:** Use of substandard bitumen or improper compaction in patch-works.
- **Lack of Inventory:** No updated "Road Management System" (RMS) to track the health of rural roads.

c. Recommended Measures:

- **Preventive Maintenance:** Shifting focus to "Early Intervention" to extend pavement life.

- **Performance-Based Contracts (PBC):** Making contractors responsible for maintaining the road quality for a fixed period.
- **Mechanized Maintenance:** Using pothole-patching machines instead of manual labor for better quality.
- **Strengthening RBN:** Increasing fuel cess or toll fees to ensure a dedicated maintenance fund.

Section-D (25 Marks)

10. What is Biological Oxygen Demand? Why is the important in designing wastewater systems? [5]

Answer:

i. **Definition:**

BOD is the amount of dissolved oxygen (DO) required by aerobic microorganisms to stabilize biodegradable organic matter in water at a specific temperature ($20°C$) over a specific period (usually 5 days, BOD_5).

ii. **Importance in Wastewater Design:**

- **Pollution Strength:** It indicates the "Organic Loading" of the sewage. Higher BOD means more concentrated waste.
- **Treatment Selection:** Determines whether a "Primary" (physical) or "Secondary" (biological like ASP or Reed Bed) treatment is necessary.
- **Effort and Sizing:** Used to size Aeration Tanks and Trickling Filters.
- **Discharge Compliance:** Nepal's "Generic Standards" limit BOD discharge into inland surface waters (usually < 50 mg/L).
- **Efficiency Monitoring:** By comparing Influent BOD and Effluent BOD, the plant's performance is evaluated.

11. Describe various step involved in engineering, design, cost estimate and detailed study of a rural water supply system? [10]

Answer:

A typical Rural Water Supply System (RWSS) in Nepal follows these systematic steps:

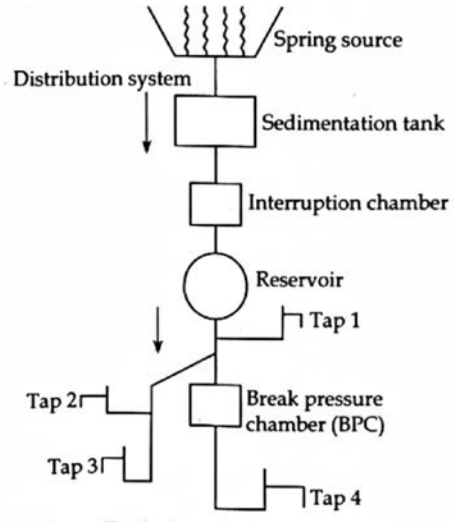


Figure: Typical rural water supply system

i. **Project Identification and Pre-feasibility:** Site visit, interaction with the community, and identifying potential sources (Springs/Streams).

ii. **Topographic Survey:** Using GPS/Total Station to map the source, storage tank (RVT) locations, and distribution clusters.

iii. **Hydrological Study:** Measuring the "Safe Yield" of the source during the dry season (Pre-monsoon).

iv. **Socio-economic Survey:** Determining the current population, growth rate (usually 1% to 2.1%), and "Willingness to Pay."

v. **Engineering Design:**

- **Demand Projection:** Calculating the "Base Year" and "Design Year" (usually 15-20 years) water demand.
- **Hydraulic Analysis:** Using software like EPANET to size pipes, ensuring residual pressure (min 5-10m or as per standards) at all taps.

vi. **Structural Design:** Designing the Intake, Collection Chamber, Break Pressure Tanks (BPT), and RVT.
vii. **Cost Estimation:** Preparing the Bill of Quantities (BoQ) based on current "District Rates" and adding VAT and contingencies.
viii. **Environmental Study:** Initial Environmental Examination (IEE) to assess impacts.
ix. **Reporting:** Preparing the Detailed Project Report (DPR) for government approval.

12. **What are the key indicators mentioned in Sustainable Development Goals 6 related to water supply and sanitation? How can Nepal achieve the goal? [5+5=10]**

Answer:

a. Sustainable Development Goal 6 (SDG 6) aims to "ensure availability and sustainable management of water and sanitation for all" by 2030. Key SDG 6 Indicators:

- **Indicator 6.1.1:** Proportion of population using safely managed drinking water services (accessible on premises, available when needed, and free from contamination).
- **Indicator 6.2.1:** Proportion of population using safely managed sanitation services, including a hand-washing facility with soap and water.
- **Indicator 6.3.1:** Proportion of wastewater safely treated.
- **Indicator 6.b.1:** Proportion of local administrative units with established and operational policies and procedures for participation of local communities in water and sanitation management.

b. How Nepal Can Achieve the Goal:

- **Focus on Quality:** Moving from "Basic Coverage" to "Safely Managed" by installing water treatment plants (WTP) in existing systems.
- **Climate Resilience:** Designing "Climate Resilient" intakes that can withstand floods and droughts.
- **Institutional Strengthening:** Empowering Local Governments (Palikas) to manage water utilities effectively under the federal structure.
- **Fecal Sludge Management (FSM):** Moving beyond "Open Defecation Free" (ODF) status to the safe treatment and disposal of waste from septic tanks.
- **Data Driven Planning:** Implementing "Water Safety Plans" (WSP) in every community-managed scheme.
- **Securing Sustainable Financing:** Increased investment in maintenance and rehabilitation, along with fostering public-private partnerships, is needed to meet the financial requirements for achieving SDG 6 targets